

Dr. Joseph Karianjahi Njeri

Full-Stack Data Scientist | Analytics Engineer | Backend Developer

Python | Django | REST APIs | PostgreSQL | Machine Learning | JavaScript | Cloud

Contact Information

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GitHub	github.com/karianjahi
Portfolio	karianjahi.github.io/personal-portfolio

Professional Profile

Full-Stack Data Scientist and Software Developer with a PhD (Dr. rer. nat.) and several years of professional experience in data science, analytics, backend development, and scientific software engineering.

Specialized in building end-to-end analytics applications using **Python, Django, Django REST Framework, PostgreSQL, pandas, scikit-learn, and JavaScript**.

Experienced in designing production-ready systems for data ingestion, validation, machine learning, KPI computation, REST API development, interactive dashboards, and cloud deployment.

Open to roles as **Data Scientist, Analytics Engineer, Full-Stack Developer, Backend Developer**, or hybrid positions combining software engineering and advanced analytics.

Core Competencies

- **Backend Development:** Python, Django, Django REST Framework, REST APIs
- **Data Science:** pandas, NumPy, scikit-learn, Machine Learning, Statistical Modeling
- **Databases:** PostgreSQL, SQL, Data Modeling
- **Frontend Development:** JavaScript, HTML, CSS, Chart.js
- **Deployment:** Railway, Render, Linux, Gunicorn, WhiteNoise, Environment Variables
- **Software Engineering:** Clean Code, Git, Debugging, Refactoring, Testing
- **Cloud Technologies:** AWS fundamentals, Docker, CI/CD concepts

Selected Projects

• Marketing Analytics Dashboard

Developed a full-stack marketing analytics platform for uploading, validating, analyzing, and visualizing campaign performance data.

- Implemented CSV upload with validation, duplicate detection, and upload audit logging
- Built KPI calculations including CTR, CPC, CPA, ROAS, and Conversion Rate
- Developed dynamic filtering, global search, sorting, and server-side pagination
- Created interactive dashboards using Chart.js
- Added CSV export for filtered datasets
- Designed REST APIs for data ingestion, KPI summaries, chart data, and reporting
- Deployed the application to Railway using PostgreSQL and environment variables

Technologies: Django, Django REST Framework, PostgreSQL, pandas, JavaScript, Chart.js, Railway

GitHub: github.com/karianjahi/marketers_paradise

Project live at: marketersparadise-production.up.railway.app

• AppPulse – Customer Behaviour Analytics for Jira App Vendors

Designed and developed a production-style SaaS analytics platform for monitoring customer health, product adoption, churn risk, and retention for Atlassian Marketplace applications.

- Built browser-based CSV upload and ingestion pipeline with schema validation, mandatory `installed_at` field, duplicate detection, and upload audit logging
- Developed rule-based customer health scoring using usage, feature adoption, reliability, and support metrics
- Implemented churn-risk prediction using Logistic Regression with scikit-learn and persisted models using joblib
- Created historical health snapshots for trend analysis and deterioration monitoring
- Added automated alerts, narrative insights, and short-term forecasting of health scores
- Built interactive dashboards with KPI cards, health distribution charts, feature importance, and risk rankings
- Developed customer detail pages with health metrics, risk explanations, recommendations, recent events, and historical charts
- Designed REST APIs for customers, events, customer health, ML metrics, feature importance, and health history
- Implemented management commands to update customer health and retrain machine learning models
- Deployed the application to Railway using PostgreSQL, Gunicorn, WhiteNoise, and environment variables

Technologies: Django, Django REST Framework, PostgreSQL, pandas, scikit-learn, joblib, JavaScript, Chart.js, Gunicorn, WhiteNoise, Railway

GitHub: github.com/karianjahi/atlassian-app-analytics

Project Live at: customerhealth-atlassian-app-analytics.up.railway.app

Professional Experience

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| 11/2021 – 07/2025 | Data Science Coach / Software Engineering Coach, Spiced Academy / neufische GmbH, Berlin, Germany <ul style="list-style-type: none">• Taught Python, SQL, Machine Learning, APIs, and software engineering• Mentored hands-on full-stack and data science projects• Supported learners with architecture design, debugging, and deployment• Introduced best practices in Git, testing, and clean code |
| 07/2019 – 06/2021 | Data Scientist, Engineering Data Intelligence (EDI) GmbH, Karlsruhe, Germany <ul style="list-style-type: none">• Developed data-driven applications and analytical pipelines• Processed and modeled large datasets• Built statistical and machine learning models• Delivered visualizations and decision-support tools |
| 05/2016 – 06/2019 | Research Associate / Research Software Developer, Karlsruhe Institute of Technology (KIT), Germany <ul style="list-style-type: none">• Developed scientific software for data analysis and modeling• Processed large meteorological datasets• Automated complex computational workflows |
| 01/2013 – 06/2015 | Lecturer, South Eastern Kenya University, Kenya <ul style="list-style-type: none">• Taught meteorology and environmental sciences• Supervised student projects |

Education

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| 10/2015 – 05/2019 | Dr. rer. nat. (PhD) in Meteorology , Karlsruhe Institute of Technology (KIT), Germany
Dissertation: Synoptic-scale dynamics and modeling of extreme precipitation in Kenya. |
| 06/2021 – 09/2021 | Data Science and Machine Learning Bootcamp , Spiced Academy, Berlin, Germany |
| 10/2010 – 08/2012 | M.Sc. in Meteorology , University of Nairobi, Kenya |
| 09/2023 – Expected 2026 | Vocational Training as Surveying Technician , Heinrich-Hübsch-Schule, Karlsruhe, Germany |

Languages

- English – Full Professional Proficiency (C2)
- German – Intermediate (B1/B2, continuously improving)
- Swahili – Native

Awards

- DAAD Scholarship

Professional Interests

Data Science, Analytics Engineering, Machine Learning, SaaS Analytics, Business Intelligence, Backend Development, Full-Stack Web Development, and Cloud Technologies.